



STUDENT DETAILS

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Unit Code: BIT4139

Unit Name: Ethical Hacking and Counter Measures

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Academic Year: 4.2

PROJECT DETAILS

Project Title: Steps In Ethical Hacking

Selected Technologies

- Nmap/ Zenmap
- DNSDumpster
- Wappalyzer
- Whois

Online Portfolio: [Github Link](#)

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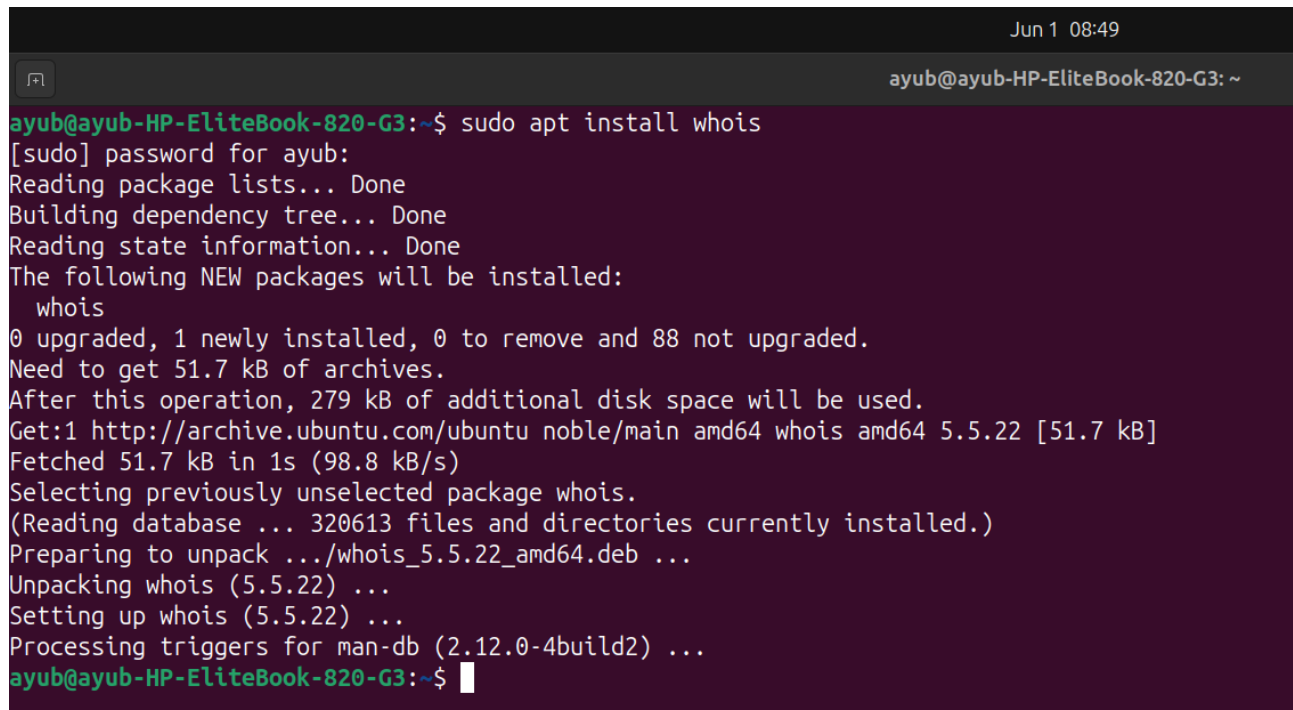
Task 1: Information Gathering

Target Identification

Selected Organization: Machakos Institute of Technology (MIT)

Website: <https://www.mit.ac.ke/>

Domain Information



```
Jun 1 08:49
ayub@ayub-HP-EliteBook-820-G3: ~
ayub@ayub-HP-EliteBook-820-G3:~$ sudo apt install whois
[sudo] password for ayub:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  whois
0 upgraded, 1 newly installed, 0 to remove and 88 not upgraded.
Need to get 51.7 kB of archives.
After this operation, 279 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 whois amd64 5.5.22 [51.7 kB]
Fetched 51.7 kB in 1s (98.8 kB/s)
Selecting previously unselected package whois.
(Reading database ... 320613 files and directories currently installed.)
Preparing to unpack .../whois_5.5.22_amd64.deb ...
Unpacking whois (5.5.22) ...
Setting up whois (5.5.22) ...
Processing triggers for man-db (2.12.0-4build2) ...
ayub@ayub-HP-EliteBook-820-G3:~$
```

Figure 1: Installation of WHOis

```
ayub@ayub-HP-EliteBook-820-G3:~$ whois mit.ac.ke
Domain Name: mit.ac.ke
Registry Domain ID: 12326-Kenic
Registrar WHOIS Server: https://hostafrika.com
Registrar URL: https://hostafrika.com
Updated Date: 2026-01-15T08:09:52Z
Creation Date: 2008-02-05T21:00:00Z
Registry Expiry Date: 2027-02-05T21:00:00Z
Registrar Registration Expiration Date: 2027-02-05T21:00:00Z
Registrar: HOSTAFRICA EAC
Registrar Street Address: NEXTGEN MALL 5th FLOOR Office NO. 14
MOMBASA ROAD NAIROBI
P.O. Box 3543 - 00200 Nairobi
Registrar City: Nairobi
Registrar State: Nairobi
Registrar Postal Code: 001000
Registrar Phone: 0709399000
Registrar Email: noreply@hostafrika.com
Registrar Abuse Contact Email: support@hostafrika.com
Registrar Abuse Contact Phone:
Domain Status: active https://icann.org/epp#active
Domain Status: serverRenewProhibited https://icann.org/epp#serverRenewProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Name Server: ns2.host-ww.net
Name Server: ns1.host-ww.net
DNSSEC: unsigned
>>> Last update of WHOIS database: 2026-06-01T05:51:24.733Z <<<

For more information on domain status codes, please visit https://icann.org/epp
```

Figure 2: Obtain Domain Information Using WHOis

IP Address Identification

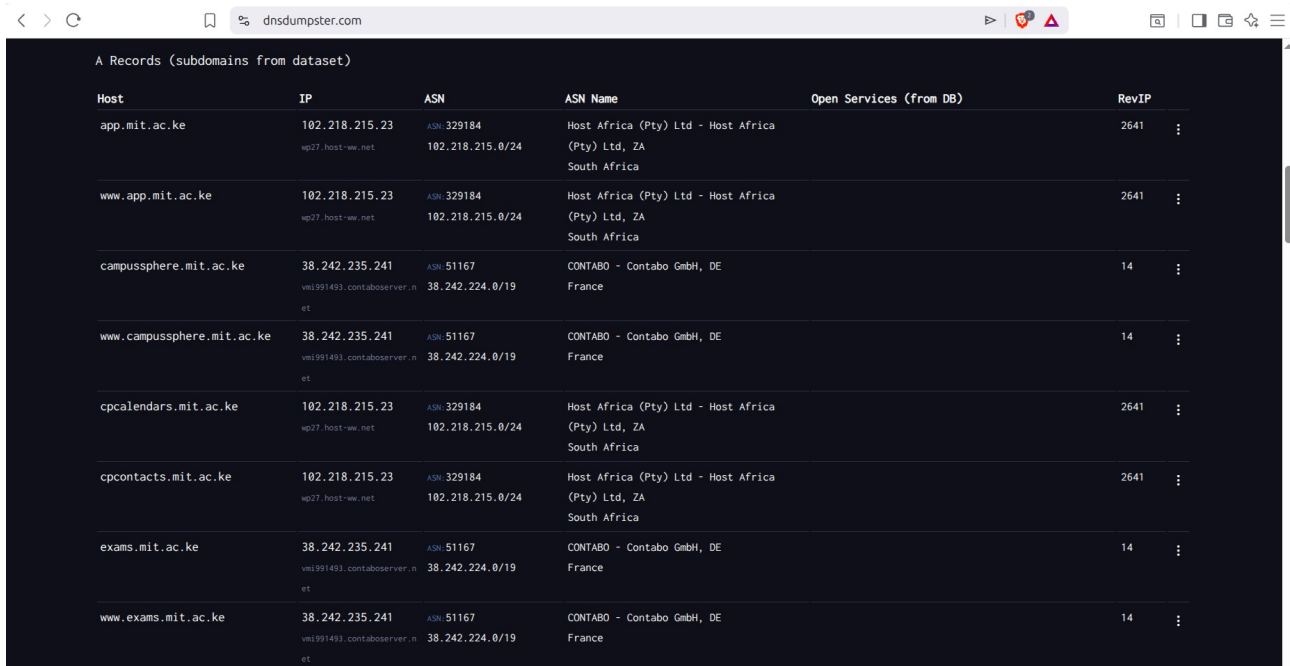
```
ayub@ayub-HP-EliteBook-820-G3:~$ nslookup mit.ac.ke
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   mit.ac.ke
Address: 102.218.215.23
```

Figure 3: IP Address Identification

Subdomains

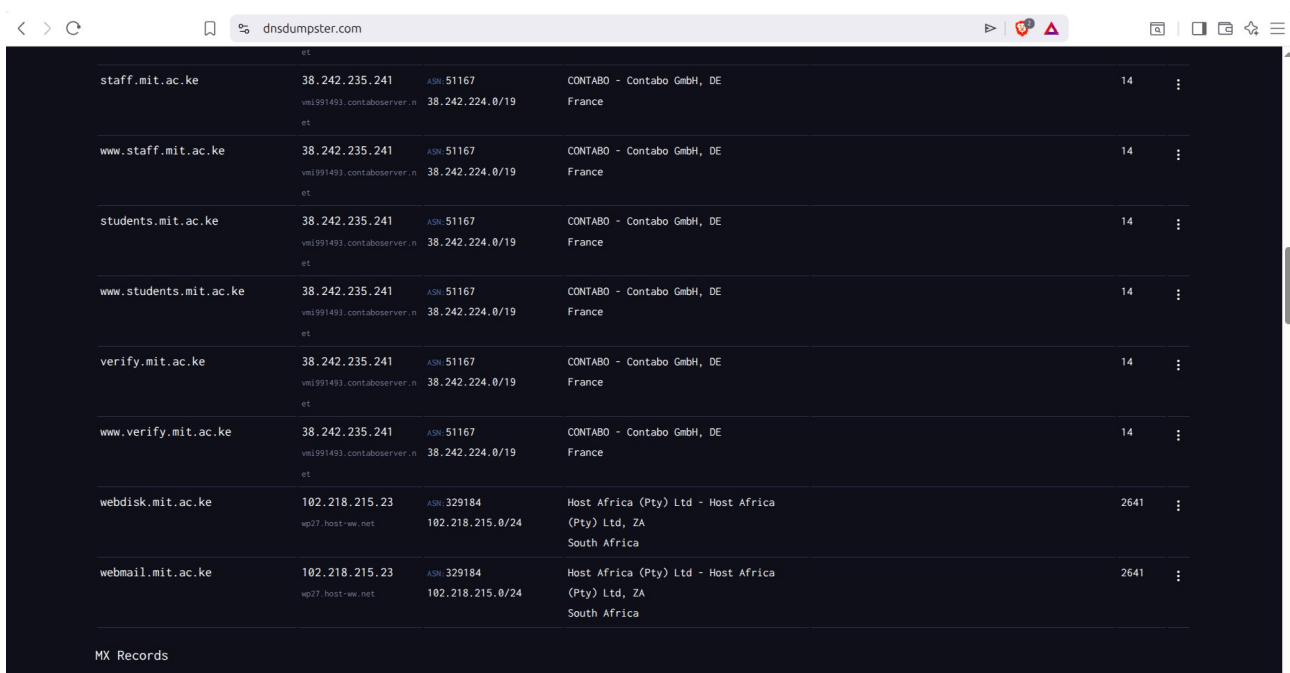
A site called DNSDumpster was used to get a list of all subdomains present



A Records (subdomains from dataset)

Host	IP	ASN	ASN Name	Open Services (from DB)	RevIP
app.mit.ac.ke	102.218.215.23 wp27.host-ew.net	ASN: 329184 102.218.215.0/24	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA South Africa		2641
www.app.mit.ac.ke	102.218.215.23 wp27.host-ew.net	ASN: 329184 102.218.215.0/24	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA South Africa		2641
campussphere.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
www.campussphere.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
cpcalendars.mit.ac.ke	102.218.215.23 wp27.host-ew.net	ASN: 329184 102.218.215.0/24	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA South Africa		2641
cpcontacts.mit.ac.ke	102.218.215.23 wp27.host-ew.net	ASN: 329184 102.218.215.0/24	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA South Africa		2641
exams.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
www.exams.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14

Figure 4: Subdomain List 1



staff.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
www.staff.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
students.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
www.students.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
verify.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
www.verify.mit.ac.ke	38.242.235.241 vms991493.contaboserver.net	ASN: 51167 38.242.224.0/19	CONTABO - Contabo GmbH, DE France		14
webdisk.mit.ac.ke	102.218.215.23 wp27.host-ew.net	ASN: 329184 102.218.215.0/24	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA South Africa		2641
webmail.mit.ac.ke	102.218.215.23 wp27.host-ew.net	ASN: 329184 102.218.215.0/24	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA South Africa		2641

MX Records

Figure 5: Subdomain List 2

webmail.mit.ac.ke	102.218.215.23	ASN: 329184	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA	2641	
	wp27.host-ww.net	102.218.215.0/24	South Africa		
MX Records					
0 mit.ac.ke	102.218.215.23	ASN: 329184	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA		
	wp27.host-ww.net	102.218.215.0/24	South Africa		
NS Records					
ns1.host-ww.net	169.239.180.4	ASN: 329184	Host Africa (Pty) Ltd - Host Africa (Pty) Ltd, ZA	http: unknown server	
	ns1.host-ww.net	169.239.180.0/24	South Africa	title: Not Found	
				https: unknown server	
				title: Not Found	
				cn: ns1.host-ww.net	
ns2.host-ww.net	195.110.43.170	ASN: 8648	ONE-NETWORK - dogado GmbH, DE		
	ns2.host-ww.net	195.110.42.0/23	Germany		
TXT Records					
"v=spf1 +a +mx +ip4:102.218.215.23 include:spf.antispamcloud.com include:spf.host-ww.net +ip4:102.218.215.34 +ip4:136.243.150.25 -all"					

Figure 6: Subdomain List 3

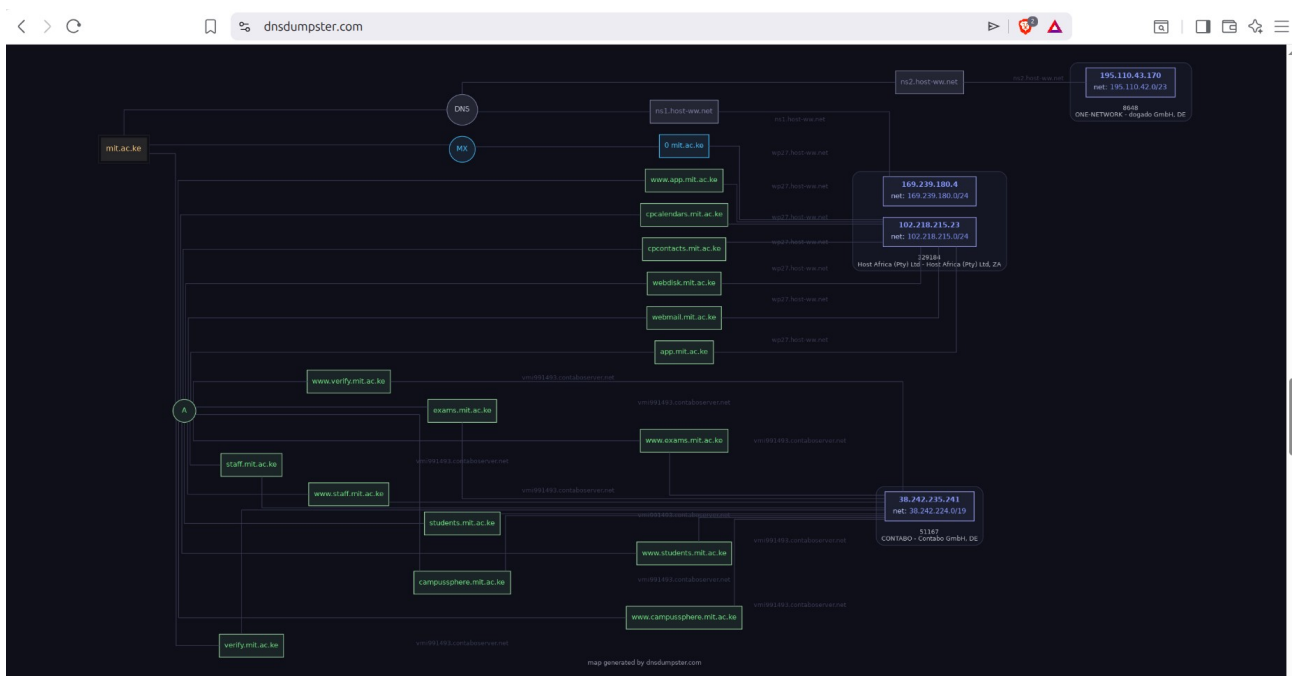


Figure 7: Mapping of all interrelations

Technology Used

Technology stack

CMS
 WordPress
Blogs
 WordPress
Programming languages
 PHP (7.4.33)
Databases
 MySQL
Web servers
 LiteSpeed

Figure 8: Wappalyzer screen showing all technologies used for the website

Locale
Countries Kenya
Security Certificate protocol TLS 1.3 Certificate expiry Jul 16, 2026 SPF record  DMARC record 

Figure 9: Wappalyzer image showing more details on technology used

Open Services Enumeration

You can do the following without logging in:

- View home page
- View list of courses
- Access enrollment section and download application form
- View images of previous events
- Get links to respective student or staff portals
- Obtain contact information

Task 2: Scanning and Enumeration

Scan Ports

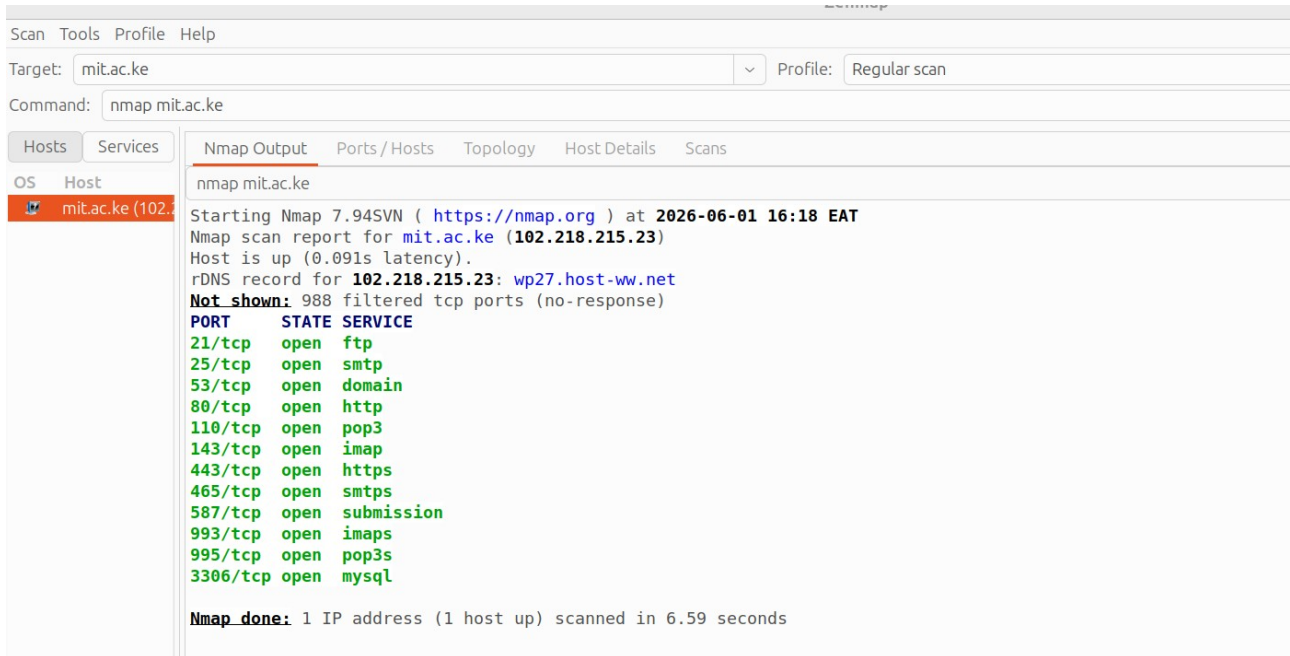


Figure 10: Regular nmap port scan

Detect Running Services

```
ayub@ayub-HP-EliteBook-820-G3:~$ nmap -sV mit.ac.ke
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-01 16:21 EAT
Nmap scan report for mit.ac.ke (102.218.215.23)
Host is up (0.094s latency).
rDNS record for 102.218.215.23: wp27.host-ww.net
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      Pure-FTPd
25/tcp    open  smtp?
53/tcp    open  domain   PowerDNS Authoritative Server 4.9.5
80/tcp    open  http     LiteSpeed httpd
110/tcp   open  pop3     Dovecot pop3d
143/tcp   open  imap     Dovecot imapd
443/tcp   open  ssl/http LiteSpeed httpd
465/tcp   open  ssl/smtp Exim smtpd 4.99.2
587/tcp   open  smtp     Exim smtpd 4.99.2
993/tcp   open  ssl/imap Dovecot imapd
995/tcp   open  ssl/pop3 Dovecot pop3d
3306/tcp  open  mysql    MySQL 5.5.5-10.6.25-MariaDB-cll-lve

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 176.85 seconds
ayub@ayub-HP-EliteBook-820-G3:~$
```

Figure 11: Identification of running services

Identify OS Version

```
ayub@ayub-HP-EliteBook-820-G3:~$ sudo nmap -O mit.ac.ke
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-06-01 16:28 EAT
Nmap scan report for mit.ac.ke (102.218.215.23)
Host is up (0.071s latency).
rDNS record for 102.218.215.23: wp27.host-ww.net
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   open  pop3
143/tcp   open  imap
443/tcp   open  https
465/tcp   open  smtps
587/tcp   open  submission
993/tcp   open  imaps
995/tcp   open  pop3s
3306/tcp  open  mysql
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: game console|media device
Running (JUST GUESSING): Microsoft embedded (87%), Sling embedded (85%)
Aggressive OS guesses: Microsoft Xbox game console (modified, running XboxMediaCenter) (87%), Slingbox Pro-HD TV over
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.22 seconds
```

Figure 12: Attempts of identifying the version of OS used

Shared Folders

No SMB services were detected and thus shared folder enumeration could not be performed

Reflection

Through this task I gained hands on experience in scanning techniques used in penetration testing. The exercise helped me understand the importance of information gathering and the potential security risks associated with exposed services and info that is publicly available